BS-361-19-23

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Bahauddin Zakariya University Multan

Program: B.S in (Physics)

Session: 2019 -2023

Time Allowed 02:30 hrs

Course Title: Classical Mechanics (PHYS -305)

Semester: 5th (Final Term)

Maximum Marks: 60

Note : Attempt all the Questions.

Q. 1:	**Describe Briefly** i. What do you mean by Scattering? Define "Solid angle" and "Scattering angle". ii. State "Principle of Virtual Work" for static system. What is its importance? iii. What do you mean by "Calculus of Variations"? Give its significance. iv. Define "Fundamental Poisson Brackets" and give their significance? v. Describe "Cyclic Coordinate" with its example. vi. Show that kinetic energy $T = T' + T_{C.M}$ for the dynamic system of particles? vii. Discuss the difficulties produced by "CONSTRAINTS" in the solution of mechanical problems. How can these difficulties be removed? viii. What do you mean by the "Central Force"? Give few examples. ix. What is "Legendre Transformation"? x. State "Liouville's Theorem" and its importance.	2×10 = 20
	LONG QUESTIONS	
Q. 2:	(a) Determine the "Lagrange Equation of motion" for an "Atwood Machine". (b) Calculate Minimum "Surface of Revolution" by taking some curve passing between two points (x_1, y_1) and (x_2, y_2) and revolving it about y-axis by applying "Hamilton's Principle".	(5+5)
Q. 3:	(a) State Kepler's Laws of planetary motion and show that Areal Velocity in a Central Force Field is Constant. (b) Find under what condition $Q = \frac{\alpha p}{x}, P = \beta x^2$, where α and β are constants, represent a "Canonical Transformation" for a system of one degree of freedom and obtain a suitable "Generating Function".	(5+5)
Q. 4:	Discuss Rutherford Scattering to find out a relation: $$\frac{dN}{d\theta} = -\frac{N_o}{A}\frac{\mu_1^2 \pi}{v_o^4}\frac{\cos\theta/2}{\sin^3\theta/2}$$ for particles scattered through θ and θ+dθ. All parameters have their usual meanings.	(10)
Q. 5:	a) What do you know about Canonical Transformations? Find Equations of Canonical Transformation when Generating Function $F = F_1(q, Q, t)$. b) Define Poisson Bracket and verify that $[UV, W] = U[V, W] + [U, W]V$, where $U = U(q, p, t)$, $V = V(q, p, t)$ and $W = W(q, p, t)$.	(5+5)

BS-361-19-23-200.

BS-Physics Semester V (2019-23) BS-359-19-23

Paper: Mathematical Methods of Physics-1 PHY-301

Marks: 60 Phys-301 *Time: 2.30 hrs*

Q.No.1 10 X 2 =20

a) Is velocity of a fluid passing through tunnel a Tensor quantity? 2

b) Either the Eigen Values of an Anit-Hermitian Matrix are Real or Imaginary? 2

c) The set of cube root of unity form a group is it continuous group? 2

d) Differentiate between the Simply and Multiply connected Regions? 2

e) Find the divergence of **v**, where **v** = x^2 **i** – $8y^3$ **j**+ 4{ x + z^3} **k** 2

f) Find the work done Integral for the force **F** = 4xz **i** – y^2 **j** + yz **k** for the displacement d = 2x **i** - 4y **j** +7z **k** 2

g) Find ∇ X **f (r)** where f (r) = 3 xy **i** – 7yz **k** – 2xz **j**? 2

h) Diagonalize the matrix A = $\begin{matrix} -i & 0 \\ 1 & -1 \end{matrix}$? 2

i) Is Kronecker delta an Isotropic Tensor? 2

j) Is Z^2 an analytical function? 2

Q.2 (a) Derive Cauchy"s Reimann conditions in Polar form? 10

Q.No 3 State and prove Green theorem? 10

Q.4 (a) Show that eigen values of diagonal matrix consists of diagonal elements? 5

(b) Find an expression for Curl of a vector in term of curvilinear coordinates system? 5

Q.No.5 (a) State quotient rules for tensor analysis? 5

(b) define and explain Group and its applications in Physics? 5

BS-359-19-23-2020.

BS- physics.

Final TERM EXAM (Analog electronics I)

Bahauddin Zakariya University, Multan

BS-363-19-23

BS: 5th semester (2019- 2023)

Time Allowed = 2½ hrs.

Course: PHYS-309

Max. Marks = 60

<table>
<tr><th>Question Number</th><th>Question</th><th>marks</th></tr>
<tr><td>1</td><td>Chose the Most Appropriate Answer (option) from the Given FOUR CHOICES</td><td>1×10</td></tr>
<tr><td></td><td>
i. In notonization of a circuit, we, theoretically, short the output to consider
(a). Maximum potential difference (b). Maximum resistance
(c). Maximum current flow (d). Maximum potential drop

II. Schottky diodes are used in
(a). Low frequency applications (a). High frequency applications
(c). Independent of frequency (d). DC devices

iii. In full wave rectifier frequency of output is
(a). Same as input (b). Double of input
(c). Three times of input (d). Zero frequency

iv. Thermistors are
(a). Current dependent (b). Voltage dependent
(c). Temperature dependent (d). None of these

v. Amplification operation of BJT is controlled by
(a). Voltages (b). Current
(c). Temperature (d). Electric field

vi. In cutoff condition of BJT appears as
(a). Open switch (b). Closed switch
(c). Amplifier (d). Oscillator

vii. Silicon controlled rectifier (SCR) has three terminals, which are
(a). Anode, Gate, Cathode (b). Anode, Drain, Cathode
(c). Gate, Drain, Source (d). Emitter, Collector, Base

viii. Field Effect Transistor (FET) devices are
(a). Low power high frequency (b). High power high frequency
(c). High power low frequency (d). Low power low frequency

ix. Ripple factor in the output of a rectifier is (C=capacitance, R=resistance)
(a). Directly proportional to C (b). Directly proportional to R
(c). Inversely proportional to C (d). Independent of C and R

x. Frequency modulation (FM) is for
(a). Long range communication (b). Short range communication
(c). Both long range and short range (d). Tuning the amplifiers
</td><td></td></tr>
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P.T.O.

Question Number	② Question BS-363-19-23.	marks
2	Answer the Given SHORT QUESTIONS	2×10
	i. Define Electronics. ii. Define load line for Transistors. iii. What is Tunnel Diode? iv. Describe Line Regulation. v. Draw circuit of Positive Biased Clipper. vi. What are CMOS devices? vii. Describe Optical Couplers. viii. Give an example with symbol of Voltage-Controlled Devices. ix. Draw circuit for Emitter Follower npn Transistor Amplifier. x. Define Modulation.	
	Long Questions (Subjective Part)	
3	(a) Define **Semiconducting Materials** give their **classification based on doping**. (b) What are different types of **Diode Models** explain with the help of I-V **Characteristic Curves?**	4+6
4	(a) What is **Light Emitting Diode**, briefly describe the phenomenon behind the **LED** and **photodiode**. (b) Draw waveform and circuit for UJT relaxation oscillator.	5+5
5	(a) What is **Full Wave Rectifier?** Explain **Circuitry** and **Working of Bridge Rectifier** (b) Draw I-V characteristics for **CE** configuration of transistor, label its different regions and define it.	6+4

BS-363-19-23-20.

Program: BS (Physics) **Session:** (2019- 2023) 5^{th} Semester (Final)

Course Title: Quantum Mechanics – I **Course Code:** PHYS – 303

Time Allowed: 02:30 hrs **Total Marks:** 60

Questions	Marks
Q. 1: Encircle the correct answer.	10
I. The energy of photon depends on its: a) Amplitude b) Speed c) Temperature d) Pressure II. When an electron jumps from an orbit where n = 1 to n = 3, its energy in terms of the energy of the ground level (E_1) is: a) $E_1/3$ b) $E_1/9$ c) $9E_1$ d) $3E_1$ III. If two operators are commuting with each other, they will possess a _______Eigenstate. a) Non-degenerate b) Degenerate c) Orthogonal d) Simultaneous IV. The Eigen values of Hermitian operator are always_______ a) Complex b) Imaginary c) Real d) None of them V. The bra space is ________ to the ket space. a) Dual space b) Hilbert space c) Euclidean space d) All of these VI. The states corresponding to discrete spectra are called ________ states a) Bound b) Unbound c) Stationary d) Dynamic VII. If azimuthal quantum number is 3, number of values of the magnetic quantum number is a) 3 b) 6 c) 5 d) 7 VIII. Expression for z – component of angular momentum is a) $-i\hbar\frac{\partial}{\partial\varphi}$ b) $-i\hbar\frac{\partial}{\partial\theta}$ c) $i\hbar\frac{\partial}{\partial\varphi}$ d) $i\hbar\frac{\partial}{\partial\omega}$ IX. Radial equation for central potential depends on _________ quantum number a) Azimuthal b) Magnetic c) Spin d) None X. The square of angular momentum (J^2) commutes with a) Jx b) Jy c) Jz d) All of them	
Q. 2: Short Questions	10 * 3 = 30
I. Give physical interpretation of a wave function? II. What are Dirac Notations? III. What do you know about Parity Operator? IV. Prove that, $[\hat{x}, \hat{p}_x] = i\hbar$ V. Differentiate between Observables and Operators. VI. Write an expression for finding the expectation value of an Operator. VII. Write an expression for the zero-point energy of the harmonic oscillator.	

P.T.O.

Questions 8 BS-360-19-23	Marks
VIII. What do you know about Pauli's matrices IX. Write an expression for the orbital angular momentum operator $\vec{L}^2$ in spherical polar coordinates. X. What is effective potential? Write down its expression.	
	20
Long Questions	
Q. 3: Q. 2: A flux of particles (all having the same mass m and moving with the same velocity) moving from left to the right is subjected to a potential step $V(x) = \begin{cases} V(x) = 0 \quad for \quad -\infty < x < 0 \\ V(x) = V_o \quad for \quad 0 < x < \infty \end{cases}$. Find the reflection and transmission coefficients, R and T, when energy of the particles is, E>V₀. **(10)** E>V₀ **Q. 4:** Determine the Eigen function of $\hat{L}^2$ differentially and calculate first three spherical harmonics. **(5)** **Q. 5:** Consider a spinless particle of mass m which is moving in a three-dimensional potential $V(x,y,z) = \begin{cases} \frac{1}{2} m\omega^2 z^2, & 0 < x < a, 0 < y < a, \\ \infty, & elsewhere. \end{cases}$ Write down the total energy and the total wave function of this particle. **(5)**	

BS 360-19-23-200.